

Algorithm or Skill?

A Monte Carlo Simulation of Creator Strategy and Emergent Inequality on TikTok

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Code and reproducibility. All simulation code, calibration steps, and figure-generation scripts live in the companion notebook `notebooks/viral_roulette.ipynb`, alongside an installable package (`src/viral_roulette/`) and a test suite that reproduces every calibration constant quoted below. Repository: github.com/rayyanmaan/viral-roulette.

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1 Scenario description

1.1 Motivation and research question

TikTok is unusual among major platforms because early distribution does not require an existing follower base. A new video is typically routed to an initial cohort of users before any explicit social-graph signal can matter. In that setting, it is not obvious whether “content strategy” is actually decisive, or whether early exposure randomness dominates outcomes.

This project asks a narrow counterfactual question: across a distribution of simulated platform environments, does a creator’s strategy—niche consistency, trend chasing, quality investment, or random posting—produce meaningfully different follower outcomes and virality outcomes? If mean growth is similar across strategies, does strategy still determine the *mechanism* of growth?

Simulation is the right tool for two reasons. First, strategies cannot be randomly assigned to real creators in a controlled trial. Second, TikTok’s recommendation system is not publicly specified at the implementation level, so any empirical-only approach is forced to treat the platform as a black box. A calibrated simulation allows holding all non-strategy choices fixed and varying only strategy, which is the comparison the report is built around (Appendix A2).

From a talent-management perspective, the question becomes operational: should an agency advise clients to post consistently in a niche, chase trends aggressively, invest in production quality, or simply post without a coherent plan? The goal here is not to claim a universal rule for all creators, but to make the model’s implied trade-offs explicit and check whether they are statistically stable under Monte Carlo uncertainty.

1.2 Data sources and what each one contributes

This model is calibrated from four sources (Appendix A1):

BrightData TikTok user profiles (CSV; 1,000 creators). After cleaning, all 1,000 rows are retained (Appendix A1a). The follower distribution is extremely heavy-tailed (range 1 to 14,500,000), and the implied inequality is high: the Gini coefficient is $G_{\text{real}} = 0.9086$ (Appendix A1a). That value becomes an external calibration target for inequality; the simulation should reproduce heavy tails and a strongly concave Lorenz curve, even if it cannot match the extreme tail perfectly at small simulated population sizes.

TikTok popular hashtag dataset (XLSX; 1,200 videos). This dataset provides play counts, likes, and shares per video keyed by hashtag topic (Appendix A1b). After filtering for minimum topic support, 12 topics remain (humor-adjacent categories dominate). I compute topic-specific like-rate and share-rate multipliers and use them as content-topic engagement multipliers in the recommendation funnel. The cross-topic baseline like-rate is 0.1478 and the baseline share-rate is 0.0041 (Appendix A1b). These rates are scaled ($\text{LIKE_SCALE} = 35.0$, $\text{SHARE_SCALE} = 21.0$) so that an average-quality video ($q = 0.5$) has mean engagement score $E \approx 0.32$, slightly below the amplification threshold 0.35; that choice makes virality feasible but not guaranteed, which is necessary for strategy trade-offs to exist (Appendix A1b).

Google Trends fits for four TikTok-native trends. Four trends (Sea Shanty, Corn Kid, Brat Summer, Demure) are used to fit log-normal lifecycle curves, producing four (μ, σ) pairs (Appendix A1c). The trend engine samples one pair uniformly at random when a new trend spawns, replacing an arbitrary decay schedule with a data-grounded family of lifecycle shapes.

HypeAuditor 2026 benchmarks. HypeAuditor provides a realistic tier distribution (nano/micro/mid/macro/mega) and engagement benchmarks by tier (Appendix A1d). This is used to initialize creator follower counts in a way that reflects the actual creator population composition, rather than starting everyone at zero.

1.3 What the simulation measures

Each Monte Carlo run simulates $N_{\text{creators}} = 30$ creators for $T = 60$ weekly time steps (Appendix A0). Each strategy is evaluated over $N_{\text{MC}} = 1,000$ independent runs (Appendix A6–A11).

Outputs (with units and definitions):

- **Final followers** $F_i(T)$ per creator; the report often uses the *run-level mean* $\bar{F}(T)$ across 30 creators (Appendix A6).
- **Survival rate:** fraction of runs where $\bar{F}(T) > \bar{F}(0)$ (Appendix A7).
- **Viral rate:** fraction of videos with engagement score $E > \theta$, averaged within-run then across runs (Appendix A8).
- **Gini coefficient** of the within-run follower distribution across 30 creators, averaged across runs (Appendix A9).
- **Q4/Q1 viral-rate ratio:** within each strategy, runs are binned into quartiles by final followers; the ratio compares mean viral rates in top vs. bottom quartiles (Appendix A8b).
- **Engagement score** E : a weighted sum of completion, shares, likes, and skips (defined below) (Appendix A2b).

1.4 Model structure and rules

The model has three interacting components (Appendix A2): creator agents, a recommendation funnel, and a trend engine. The intent is not to reproduce TikTok’s full ranking stack, but to implement a minimal feedback loop that (i) provides baseline exposure to all creators, (ii) amplifies content that clears an engagement threshold, and (iii) allows strategies to change how often a creator aligns with trends and how much they trade quality for speed.

1.4.1 Creator agents

Each creator i is assigned a fixed strategy that sets a quality distribution and behavior parameters at initialization (Appendix A2a). Quality q_i is drawn once from a Normal distribution and clipped to $[0.05, 1.0]$. This is a deliberate choice: by keeping quality fixed within-run, the experiments isolate strategy differences from learning effects.

Each creator also has a content-topic distribution $C_i(t)$ over $K = 12$ topics (from the hashtag dataset). Trend-adapting strategies update their content vector toward the active trend topic:

$$C_i(t+1) = (1 - \eta I(t)) C_i(t) + \eta I(t) e_{\text{trend}},$$

where $I(t) \in [0, 1]$ is trend intensity and e_{trend} is the one-hot trending topic (Appendix A2a). Niche specialists have $\eta = 0$, so C_i is stable by construction; this is verified by a unit test (Appendix A3).

Strategy parameterization. The four strategies are parameterized as follows (Appendix A2a):

Strategy	μ_q	σ_q	η	τ	posting rate
Niche Specialist	0.58	0.07	0.00	0.00	2
Trend Chaser	0.30	0.18	0.55	0.90	3
Quality-Focused	0.88	0.03	0.08	0.08	1
Random Baseline	0.45	0.22	0.28	0.38	2

Quality-focused creators have high mean quality and low variance (consistency). Trend chasers have low mean quality and high variance (speed-driven volatility) but high trend sensitivity and high posting volume. Niche specialists do not pivot by definition. The random baseline is a null strategy intended to sit between structured approaches.

1.4.2 Recommendation algorithm: a two-stage funnel

Each video is shown to an initial cohort of users before any amplification decision (Appendix A2b). The initial cohort size is a Poisson draw with mean $N_{\text{test}} = 100$, which represents irreducible exposure luck.

For each user-view, four engagement events are sampled stochastically with probabilities driven by creator quality and topic multipliers (Appendix A2b). The engagement score is then computed as

$$E = 0.40 \hat{p}_{\text{complete}} + 0.25 \hat{p}_{\text{share}} + 0.20 \hat{p}_{\text{like}} - 0.15 \hat{p}_{\text{skip}},$$

where hats denote empirical rates in the initial cohort. Completion is weighted most heavily, shares next, then likes, with skips penalizing the score. These weights are architectural choices motivated by watch-time priority; they are not estimated from data, and I treat them as part of the model definition rather than a tuned parameter.

Amplification is threshold-based: if $E > \theta$ with $\theta = 0.35$, impressions expand:

$$\text{impressions} = \text{initial} \cdot (1 + \alpha(E - \theta)), \quad \alpha = 8.0,$$

otherwise the video receives no additional distribution (Appendix A2b). New followers are a fixed fraction of impressions (conversion rate = 0.02). A weekly churn term removes 0.3% of current followers, preventing unlimited growth and representing baseline audience decay (Appendix A2b).

1.4.3 Trend engine

At each step, a new trend spawns with probability $p_{\text{trend}} = 0.12$ in the main experiments (Appendix A6). When a trend spawns, its lifecycle intensity curve is sampled from the empirical (μ, σ) pairs fitted from Google Trends (Appendix A1c). For simplicity, trends do not overlap: at most one trend is active at a time.

1.5 Key modeling assumptions

The model is intentionally minimal, so assumptions matter. The most important are (Appendix A1–A2):

1. **Quality is fixed within-run.** This isolates strategy effects but prevents learning or skill acquisition.
2. **Users are not individual agents.** The model samples engagement from distributions instead of simulating social networks; it cannot represent community-driven cascades.
3. **Single active trend.** This approximates a dominant trend regime but excludes overlapping cultural moments.
4. **Threshold amplification.** This captures a feedback loop but is not meant as a literal description of TikTok’s ranking system.
5. **No creator-creator interactions.** There are no collaborations or cross-promotion dynamics; the model is closest to nano/micro creator settings.

2 Theoretical analysis

The goal of this section is to treat the simulation as a stochastic process with analyzable components, then use theory as a benchmark for what the Monte Carlo runs should do.

2.1 Virality probability: expectation, variance, and a normal approximation

In the initial cohort, each engagement component is an empirical average of Bernoulli trials. For a component $j \in \{\text{complete, share, like, skip}\}$ and n users,

$$\hat{p}_j = \frac{1}{n} \text{Binomial}(n, p_j), \quad \mathbb{E}[\hat{p}_j] = p_j, \quad \text{Var}(\hat{p}_j) = \frac{p_j(1-p_j)}{n}.$$

By linearity of expectation,

$$\mathbb{E}[E] = 0.40 p_{\text{complete}} + 0.25 p_{\text{share}} + 0.20 p_{\text{like}} - 0.15 p_{\text{skip}}.$$

If user responses are treated as independent within the initial cohort (the assumption used in Appendix A5), then

$$\text{Var}(E) = \sum_j w_j^2 \text{Var}(\hat{p}_j),$$

with weights $w_j \in \{0.40, 0.25, 0.20, 0.15\}$ (skip uses +0.15 in variance because the sign disappears).

Because $n = 100$ and E is a weighted sum of scaled binomials, Appendix A5 uses a normal approximation for the viral probability:

$$\mathbb{P}(\text{viral}) = \mathbb{P}(E > \theta) \approx 1 - \Phi\left(\frac{\theta - \mathbb{E}[E]}{\sqrt{\text{Var}(E)}}\right),$$

where Φ is the standard normal CDF (Appendix A5).

Evaluating at the strategy mean quality values yields the analytical predictions (Appendix A5):

Strategy	$q = \mu_q$	Theory $P(\text{viral})$
Niche Specialist	0.58	0.0703
Trend Chaser	0.30	≈ 0.0000
Quality-Focused	0.88	≈ 1.0000
Random Baseline	0.45	≈ 0.0000

The point of this table is not that it matches perfectly (it does not), but that it provides a crisp baseline for interpreting the Monte Carlo results in Fig. 8 (Appendix A8).

2.2 Expected follower growth rate and why it under-predicts the simulation

In a simplified zero-initial-followers world, the expected follower increment per step for creator i can be sketched as

$$\mathbb{E}[\Delta F_i] \approx r_i \cdot \mathbb{P}(\text{viral} \mid q_i) \cdot \mathbb{E}[\text{impressions} \mid \text{viral}] \cdot \rho,$$

where r_i is posting rate, $\rho = 0.02$ is conversion rate, and the impressions term comes from the amplification mapping (Appendix A2b). Appendix A5 computes strategy-level expected increments using approximations for $\mathbb{E}[E \mid E > \theta]$ and finds very small totals over 60 steps (order 10^1 to 10^2 followers) (Appendix A5).

The Monte Carlo experiments report mean final follower counts around 13,280–13,350 for all strategies (Appendix A6). This is not a numerical glitch; it is a modeling consequence of initialization. The simulation does not start creators at zero. It draws initial followers from a realistic tier distribution (dominated by nano creators with nontrivial existing follower counts) (Appendix A1d, A2d). Once follower stock is large at $t = 0$, the model's churn

and growth terms mainly govern deviations around that stock, and the run-level mean $\bar{F}(T)$ becomes relatively insensitive to strategy choices that act through virality. In other words: the theoretical calculation describes incremental growth from a small baseline, but the simulation operates in a regime where baseline follower stock and churn balance dominate the final mean.

2.3 Why follower distributions tend toward log-normality

Follower growth in the simulation is effectively multiplicative: followers carry over from one step to the next with small random increments and decrements. A common argument is that if

$$F(t+1) = F(t) (1 + \delta_t),$$

then

$$\log F(T) = \log F(0) + \sum_{t=0}^{T-1} \log(1 + \delta_t),$$

and if the $\log(1 + \delta_t)$ terms have finite variance and weak dependence, the sum is approximately normal by a CLT heuristic, making $F(T)$ approximately log-normal (Appendix A5). For a log-normal distribution with log-std $\sigma_{\log F}$, the Gini coefficient has closed form:

$$G = 2\Phi\left(\frac{\sigma_{\log F}}{\sqrt{2}}\right) - 1.$$

Appendix A9 uses this as a theoretical comparator, then checks it against simulated and real inequality.

2.4 Recovery time after a trend

Trend adaptation can be summarized by a simple decay argument. If a trend chaser drifts toward a trend topic by an amount d_0 and then the trend ends, the residual drift decays geometrically:

$$d(k) = d_0 (1 - \eta)^k.$$

Solving for the time to return below a tolerance ε gives

$$k^* = \frac{\log(\varepsilon/d_0)}{\log(1 - \eta)}.$$

With $\eta = 0.55$, $d_0 = 0.8$, and $\varepsilon = 0.05$, this yields $k^* \approx 3.5$ weeks (Appendix A5). This calculation is not a separate experiment in the final notebook, but it is useful for interpreting why the trend chaser can be resilient even when its per-video quality is low: aggressive adaptation implies fast post-trend re-alignment.

3 Implementation and correctness

3.1 Code architecture

The notebook is organized so that each modeling component is implemented once and then re-used across experiments (Appendix A2). The code is structured around four classes:

- **Creator** stores agent state (quality, content vector, posting policy) and produces videos (Appendix A2a).
- **Algorithm** implements the initial-cohort evaluation and threshold amplification (Appendix A2b).
- **TrendEngine** manages trend arrival and intensity through time (Appendix A2c).
- **Simulation** orchestrates the weekly loop and records outcomes (Appendix A2d).

Experiments call a single Monte Carlo runner, `run_monte_carlo()`, which standardizes seeds, run counts, and output storage (Appendix A2e). This matters for readability: strategy comparisons are implemented as a loop over strategy keys instead of separate scripts, which reduces the risk of silent divergence across experimental conditions.

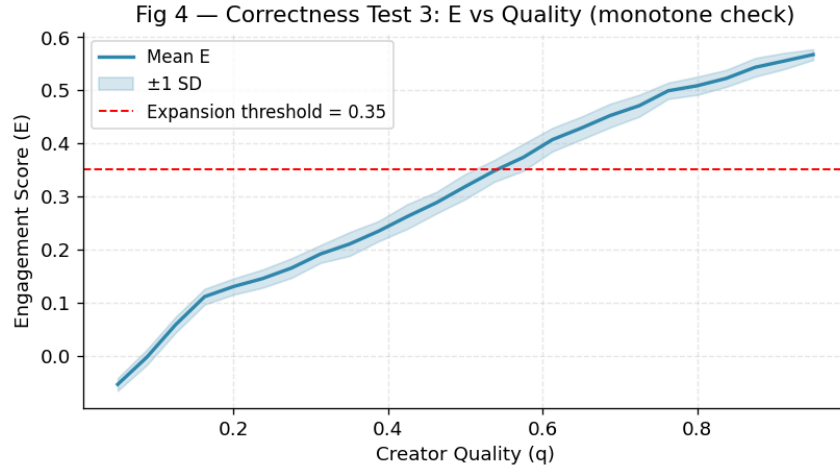


Figure 1: **Engagement score monotonicity check.** Mean engagement score increases with creator quality and crosses the amplification threshold around $q \approx 0.55$. This validates that the funnel has a stable quality signal rather than noisy or inverted behavior (Appendix A3).

3.2 Correctness tests

Appendix A3 implements five unit tests that check monotonicity, initialization, and strategy-specific invariants (Appendix A3). All five pass in the final notebook run.

The most informative test is the monotonicity check: engagement score E increases smoothly with quality, and the mean E crosses the amplification threshold near $q \approx 0.55$ (Appendix A3). This anchors interpretation later: strategies with μ_q below 0.55 should rarely go viral without a trend boost, while strategies above 0.55 should go viral regularly. The visualization from that test is included as Fig. 4.

4 Empirical analysis and results

This section builds in a specific order: first, calibration results that ground the model; then a single-run visualization that provides qualitative intuition; then the Monte Carlo experiments that answer the research question.

4.1 Calibration results

Real follower inequality. From the BrightData creator sample, $G_{\text{real}} = 0.9086$ (Appendix A1a). The Lorenz curve is extremely concave: the cumulative share of followers remains near zero for most of the creator percentiles and rises sharply near the top tail. This is the inequality target against which the simulation is compared in Experiment 4.

Topic multipliers. The hashtag dataset produces topic multipliers for like rates and share rates (Appendix A1b). Like multipliers are ordered in the plot so that longer horizontal bars indicate higher multipliers; topics like jokes and lmao are among the largest (blue), while fun and funny are among the smallest (red). Share multipliers have a different ranking: memes is the largest (around 1.3), with humor and lmao around 1.2, while fun and love are the smallest (Appendix A1b). This mismatch is useful: it prevents the model from collapsing topic choice into a single scalar “good topic” dimension.

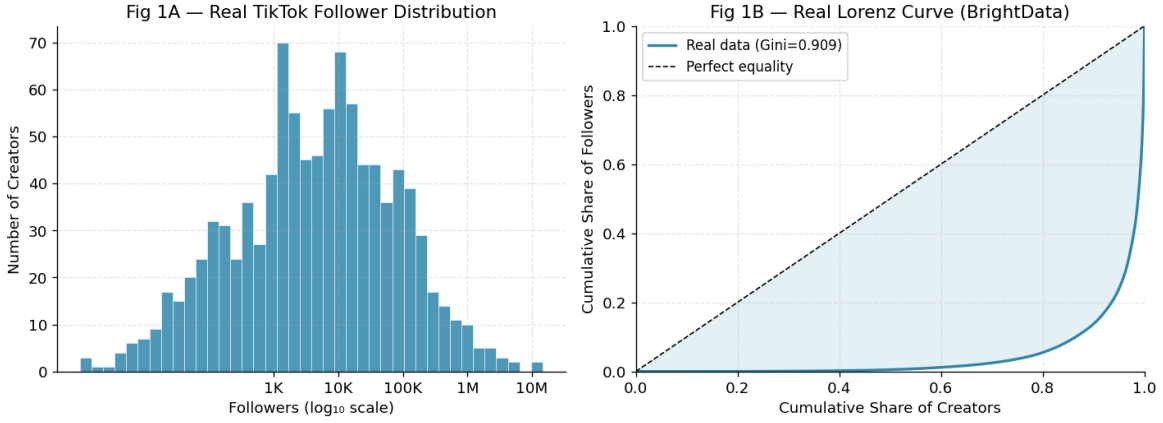


Figure 2: **Real-data calibration target.** The BrightData sample has extreme follower inequality, summarized by $G_{\text{real}} = 0.9086$. The Lorenz curve remains close to the x-axis for most percentiles, then rises sharply in the top tail (Appendix A1a).

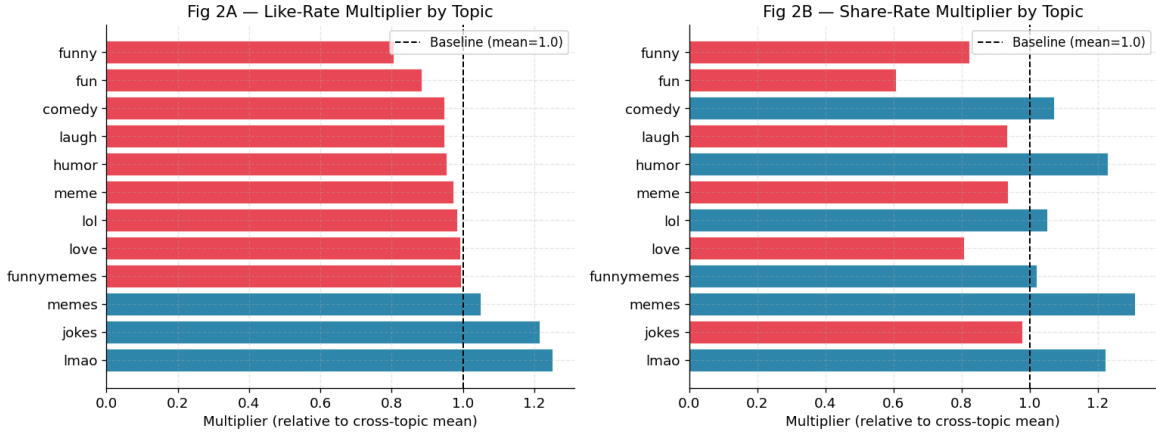


Figure 3: **Content-topic engagement multipliers.** Like-rate multipliers are visually ordered (longer bars indicate larger multipliers); share multipliers have a different topic ranking. These multipliers are used directly in the engagement probability functions (Appendix A1b).

Trend lifecycle fits. The log-normal fits track the overall trend arcs reasonably well, including Demure’s rapid rise and fall; for Sea Shanty, Corn Kid, and Brat Summer the fitted curves capture the broad pattern but miss sharp local peaks (Appendix A1c). In this model, that mismatch is acceptable because the trend engine uses the fitted curves as a family of plausible intensity schedules rather than requiring exact reconstruction of each trend.

4.2 Single-run visualization

The animation in Appendix A4 (saved as `fig5_animation.gif`) provides a sanity check for the time dynamics (Appendix A4). In the final frame at $T = 60$, all strategies decline slowly from around 5,100 followers, with the trend chaser declining slowest and thus ending highest; quality-focused is next; niche and random are below and close together. This is consistent with a regime where weekly churn is strong relative to net gains from amplification, and where the trend chaser’s higher posting volume provides more opportunities to offset churn.

Fig 3 — Log-Normal Lifecycle Fits to Real TikTok Trends (Google Trends data)

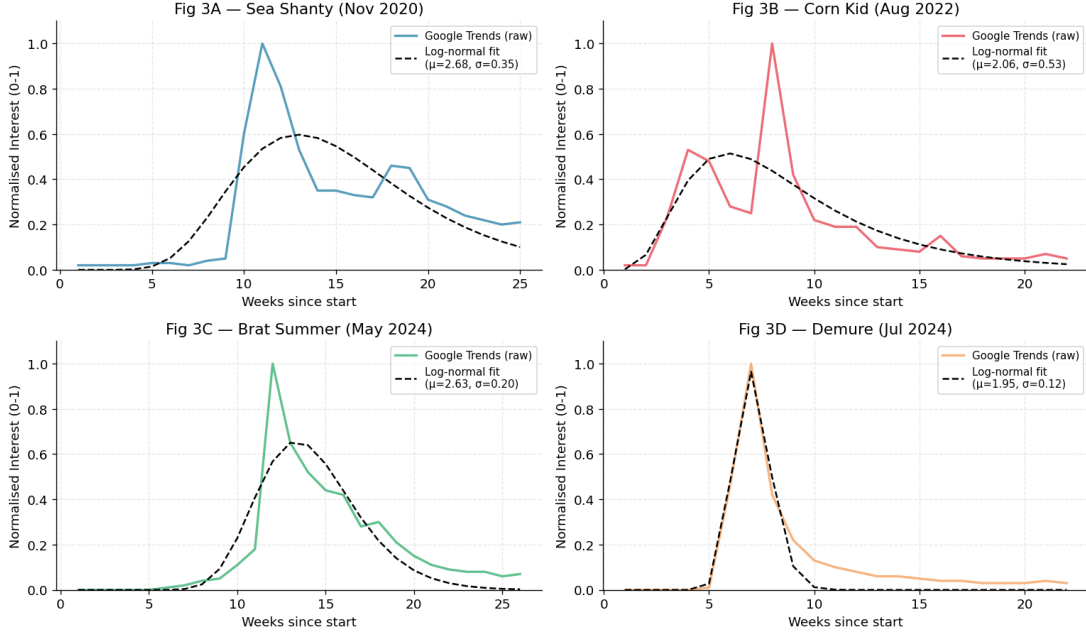


Figure 4: **Empirical trend lifecycles.** Four fitted log-normal curves provide a distribution of plausible trend shapes (flash-viral to sustained). The trend engine samples from these fitted parameter pairs when spawning new trends (Appendix A1c).

4.3 Experiment 1: follower growth distributions

Experiment 1 runs 1,000 Monte Carlo trials per strategy and records the run-level mean final followers across 30 creators (Appendix A6). The resulting histograms are nearly indistinguishable across strategies: means fall in a narrow band (about 13,281–13,351), medians cluster around 9,559–9,625, and higher moments (standard deviation and skewness) are effectively identical across strategies (Appendix A6). The confidence intervals for the mean overlap heavily (Appendix A11). This is the central null result on magnitude: in this modeled regime, strategy does not move the average final-follower metric in a meaningful way.

4.4 Experiment 2: survival rate and downside risk

To measure downside risk, I define *survival* as a run where the creator population ends with higher mean followers than it started: $\bar{F}(T) > \bar{F}(0)$ (Appendix A7). Survival rates are low for all strategies (6–11%), implying that churn dominates in most simulated environments. The ranking is nonetheless clear:

$$\text{Trend Chaser } 0.1056 > \text{Niche } 0.0735 > \text{Random } 0.0655 > \text{Quality-Focused } 0.0640,$$

with non-overlapping confidence intervals for the top-vs-bottom comparison (Appendix A7, A11).

This ordering is counterintuitive if one thinks of quality as the only driver of growth. In the model, posting rate matters mechanically because each posted video draws a new initial cohort and has an independent chance to clear the amplification threshold. The trend chaser posts three times per step (vs. one for quality-focused) and has high trend sensitivity, so even a low per-video viral probability can accumulate enough amplified impressions over time to offset churn more reliably.

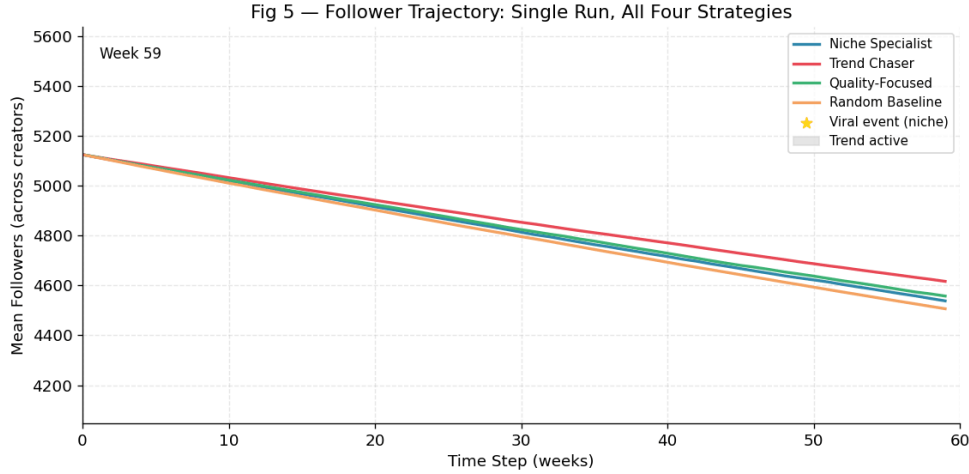


Figure 5: **Qualitative trajectory check.** In a representative single run, all strategies drift downward over 60 weeks, with the trend chaser declining slowest and ending highest. Trend windows appear as background bands in the animation (Appendix A4).

4.5 Experiment 3: virality probability (simulation vs. theory)

Virality is where strategy differentiation is strongest. Appendix A8 compares simulated viral rates to the analytical normal-approximation predictions from Appendix A5 (Appendix A8). Simulated viral rates are:

Quality-Focused 1.0000, Niche 0.3990, Random 0.1980, Trend Chaser 0.0495

(Appendix A8). These are cleanly separated by confidence intervals (Appendix A11). The theory predicts 1.0 for quality-focused and near-zero for trend chaser and random, with niche at 0.0703 (Appendix A5). The quality-focused match is exact; the others deviate upward, especially niche and random.

The deviation is explained by conditioning on trend alignment. The analytical model treats engagement probabilities as if they were evaluated at a fixed quality and topic baseline. In the simulation, when a trend is active and a creator aligns with it, share probability receives a multiplicative boost, shifting the engagement-score distribution upward and pushing many borderline cases above threshold (Appendix A2b, A8). That mechanism is strongest for mid-quality strategies (niche and random), and weaker for very high quality (already above threshold) or very low quality (still often below threshold).

4.6 Follow-up: viral rate by growth quartile

Appendix A8b asks whether virality is the mechanism separating high-growth runs from low-growth runs *within* a strategy (Appendix A8b). Runs are binned into quartiles by final followers (Q1 lowest to Q4 highest), and mean viral rates are computed per quartile.

The result is flat: within each strategy, viral rates are nearly constant across growth quartiles, and the Q4/Q1 ratios are essentially 1.00–1.01 (Appendix A8b). This matters because it rules out a tempting story: “the top runs are the ones where more videos went viral.” In this simulation, that is not the driver of run-to-run variation. Instead, run-level growth quartiles are dominated by differences in initial follower stocks produced by the tier-based initialization (Appendix A1d, A2d).

Fig 6 — Distribution of Mean Final Followers by Strategy (N=1,000 runs each)

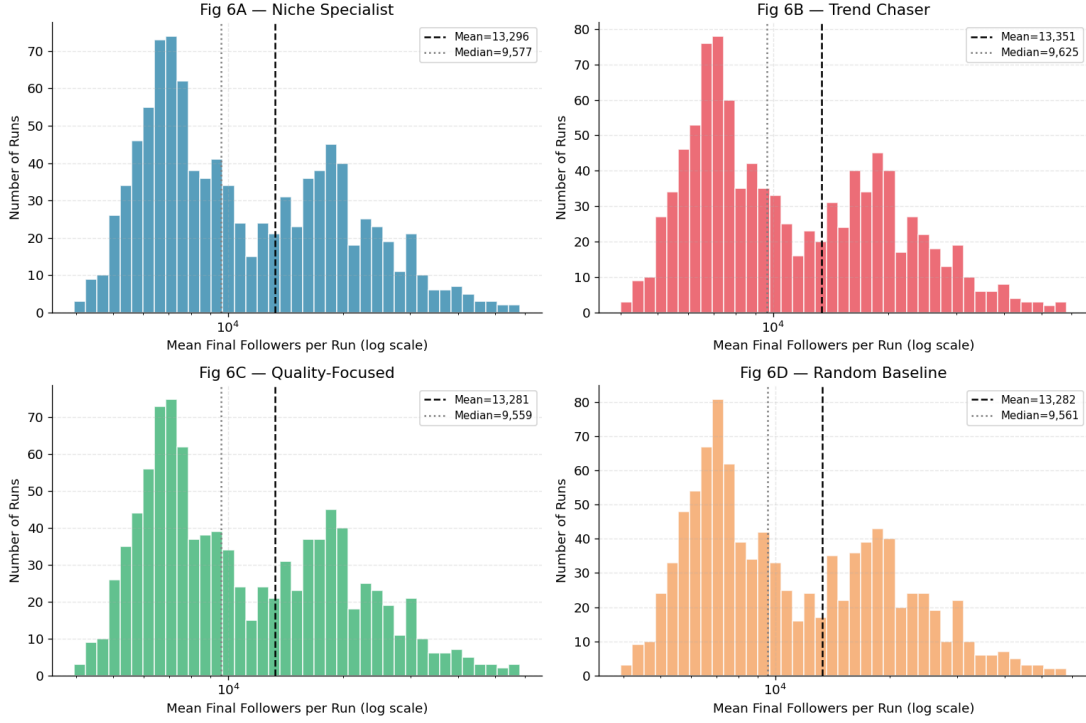


Figure 6: **Follower outcome distributions are nearly identical.** Across 1,000 runs per strategy, the four histograms align closely in center and spread. This supports the claim that in this model, mean follower magnitude is dominated by initialization and churn dynamics rather than strategy (Appendix A6, A11).

4.7 Experiment 4: inequality (Lorenz curves and Gini)

Experiment 4 computes inequality within each run (across 30 creators) and compares simulated Lorenz curves to the real-data Lorenz curve (Appendix A9). The results show three consistent patterns:

1. $G_{\text{sim}} \approx 0.555$ to 0.558 for all strategies, essentially indistinguishable across strategies (Appendix A9).
2. $G_{\text{sim}} > G_{\text{theory}} \approx 0.319$ because the simulation inherits variance from the initial follower distribution in addition to multiplicative growth variance (Appendix A5, A9).
3. $G_{\text{sim}} \ll G_{\text{real}} = 0.9086$. With only 30 creators per run, the simulation cannot represent the extreme top tail concentration visible in the 1,000-creator BrightData sample (Appendix A1a, A9).

4.8 Experiment 5: sensitivity analysis

The sensitivity analysis varies trend frequency $p_{\text{trend}} \in [0.02, 0.30]$ and quality noise $\sigma_q \in [0.05, 0.35]$ on a 5×5 grid for two strategies (trend chaser and niche), using 30 runs per cell and a smaller simulation (10 creators, 30 steps) (Appendix A10). The output heatmaps show modest variation over the grid; both strategies produce similar spatial patterns, and the difference map contains both red and blue regions, indicating that neither strategy dominates uniformly across parameter regimes (Appendix A10).

Two points are worth being explicit about. First, the overall color intensity in both heatmaps is fairly light: across the tested ranges, mean final followers do not swing wildly. Second, the two heatmaps have visually similar patterning (dark blocks appear in the same locations), which suggests that the tested parameter variations shift both strategies in the same direction over much of the grid. The difference heatmap is therefore the meaningful

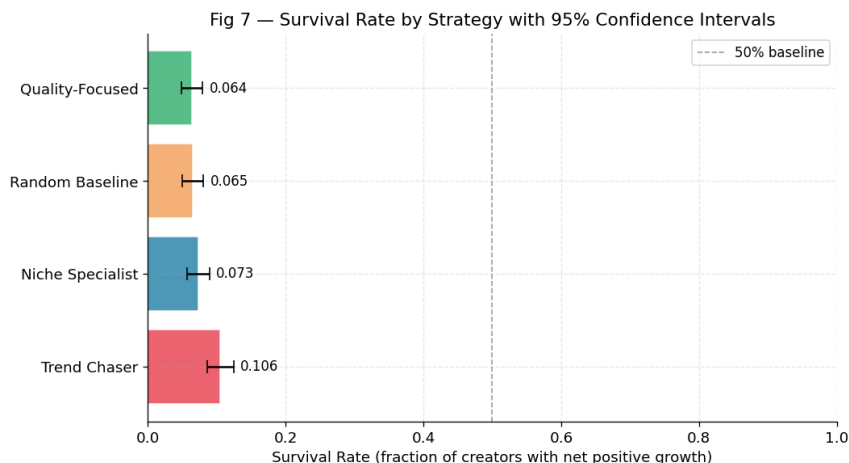


Figure 7: **Survival rate differs by strategy even when mean followers do not.** The trend chaser has the highest probability of net-positive follower change over 60 weeks, consistent with a volume-driven mechanism offsetting churn (Appendix A7).

diagnostic: it isolates where trend alignment and posting volume meaningfully change outcomes relative to niche consistency.

4.9 Confidence intervals as a result, not a footnote

Appendix A11 aggregates 95% confidence intervals across key metrics (Appendix A11). The CI summary supports three conclusions:

1. **Mean final followers are tied.** CI widths are about 1,101 followers and overlap heavily across strategies, so any ranking by mean final followers is not stable at $N = 1,000$ runs.
2. **Survival rate differences are real but small.** Trend chaser’s survival advantage over quality-focused is statistically clear because the CIs do not overlap.
3. **Viral rate is the clearest differentiator.** Viral-rate CIs are tight and non-overlapping across the four strategies; quality-focused is effectively pinned at 1.0.

5 Discussion and executive summary

5.1 Integrated interpretation

The results separate cleanly into two categories: magnitude and mechanism. On magnitude, mean final followers are essentially unchanged across strategies (Appendix A6, A11). That is not a failure of plotting or insufficient Monte Carlo repetitions; it is the outcome of modeling choices that were made deliberately for realism. Creators start with a nontrivial follower stock drawn from tier distributions, and the churn term operates on that stock. In this regime, mean final followers are driven primarily by initialization and churn balance.

On mechanism, strategies differ sharply. Viral rate is highest for quality-focused creators (1.0), then niche (0.3990), then random (0.1980), then trend chaser (0.0495) (Appendix A8). Survival rate, however, ranks differently: trend chasing has the highest probability of net-positive mean follower change over 60 weeks (Appendix A7). The model implies two distinct ways to “do well” depending on the objective: maximize per-video viral probability (quality investment) or maximize the chance of not losing ground under churn (high posting volume with trend alignment).

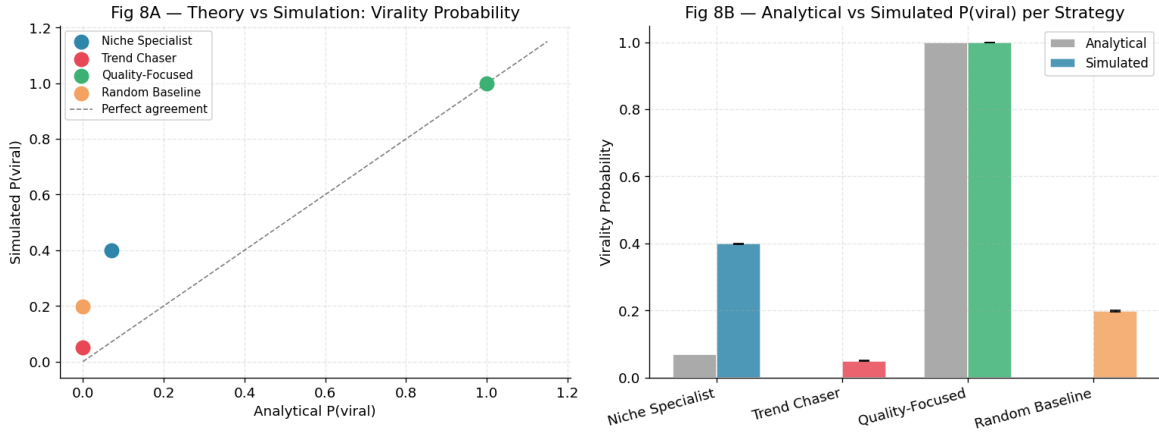


Figure 8: **Virality is strategy-dependent and theory is a partial benchmark.** All strategies lie above the theory diagonal except quality-focused, which sits at (1,1). Mid-quality strategies show the largest positive deviation because trend alignment boosts engagement in a way the independent normal approximation does not capture (Appendix A5, A8).

Inequality is high but strategy-independent in the model: the simulated Gini is around 0.556 for all strategies (Appendix A9). The gap to the real-data Gini (0.9086) is best interpreted as a scale limitation. With only 30 creators per run, the simulation cannot express the extreme right tail concentration seen on the real platform. A larger agent population is the most direct path to testing whether the model’s feedback loop can generate ultra-heavy tails.

5.2 Executive summary (non-technical reader)

This simulation compares four creator strategies over many randomized platform environments. On average, all four strategies end up with similar follower totals after 60 weeks. In the model, TikTok’s baseline distribution gives every video a chance to reach a first audience, so no strategy has a decisive advantage on total followers alone.

Where strategies differ is *how* they grow and how risky they are. A quality-focused creator goes “viral” on almost every post in this model, meaning each post earns amplified distribution. A trend-chasing creator rarely has any single post go viral, but posts more often and gets periodic boosts when aligned with a trend; that volume makes them most likely to end the year with more followers than they started with. For a client who wants reliability and wants to avoid net losses, the model favors a trend-aware, higher-volume schedule. For a client aiming for viral reach per post, the model favors quality investment.

5.3 Limitations and future work

Three limitations are directly tied to the results:

1. **No learning or skill acquisition.** Quality is fixed within-run, so the model cannot represent creators improving over time. Adding a slow quality update rule would test whether quality-focused remains dominant on virality when learning is possible.
2. **No user network structure.** Users are sampled from a static preference distribution, so community-driven cascades are absent. Introducing even a simple community graph would likely raise inequality and could specifically advantage niche specialists.
3. **Small creator population per run.** With 30 creators, the Lorenz curve cannot develop the extreme top tail seen in real data. Scaling to 500–1,000 creators per run would be computationally heavier but would give a cleaner inequality comparison.

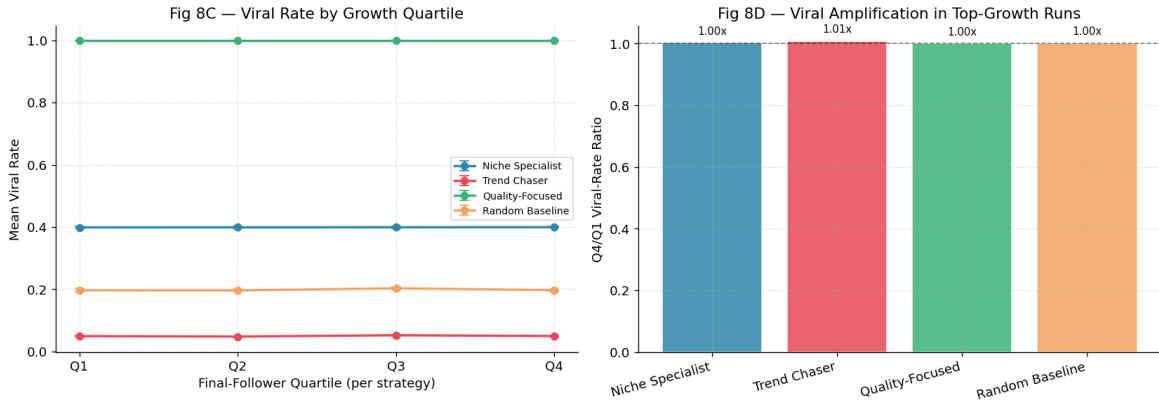


Figure 9: **Within-strategy growth is not explained by within-strategy virality.** Viral rates are nearly constant from Q1 to Q4 within each strategy, and Q4/Q1 ratios are essentially 1. This supports the claim that initialization variance dominates run-level growth differences (Appendix A8b).

6 References

References

- [1] Berry, A. C. (1941). The accuracy of the Gaussian approximation to the sum of independent variates. *Transactions of the American Mathematical Society*, 49(1), 122–136.
- [2] Bright Data. *TikTok user profiles dataset* (CSV; creator-level sample). Used for follower distribution and Lorenz/Gini calibration (Appendix A1a).
- [3] Kaggle (CC0). *TikTok popular hashtags dataset* (XLSX; video-level engagement by hashtag). Used for topic multipliers (Appendix A1b).
- [4] HypeAuditor (2026). *State of Influencer Marketing 2026*. Used for tier distribution and benchmark engagement by tier (Appendix A1d).
- [5] Google Trends. Weekly interest time series for Sea Shanty (Nov 2020–Apr 2021), Corn Kid (Aug–Dec 2022), Brat Summer (May–Oct 2024), and Demure (Jul–Nov 2024). Used for log-normal lifecycle fitting (Appendix A1c).

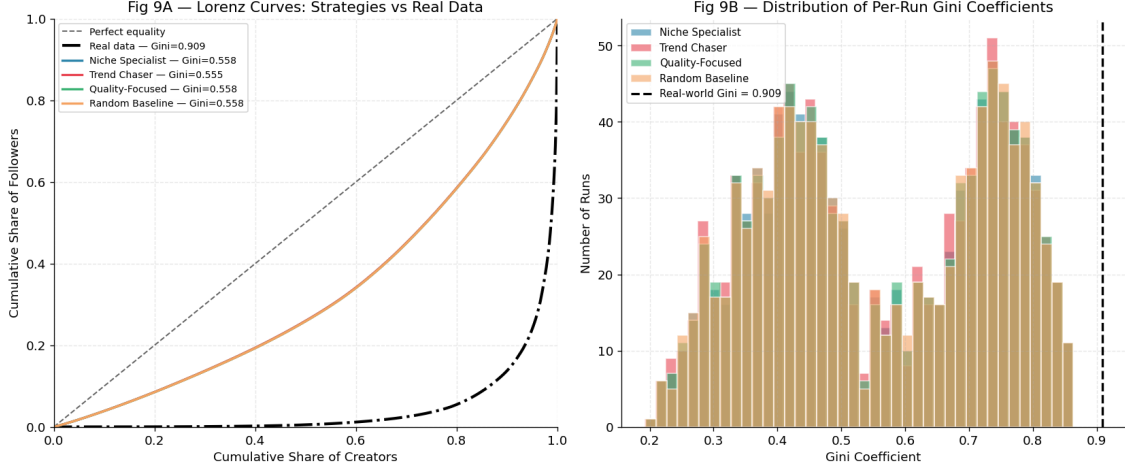


Figure 10: **Inequality is high but strategy-independent in this model.** Simulated Lorenz curves overlap across strategies and lie between the equality diagonal and the real-data curve. The model produces substantial inequality ($G_{\text{sim}} \approx 0.556$) but cannot reach the extreme concentration in the real dataset at $N = 30$ creators (Appendix A9).

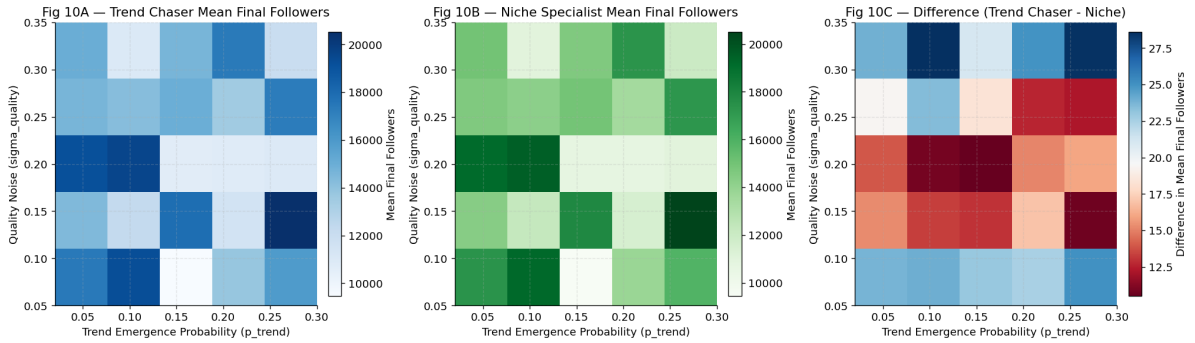


Figure 11: **Sensitivity is modest; relative performance depends on regime.** Both strategies show similar broad patterns over $(p_{\text{trend}}, \sigma_q)$, while the difference heatmap highlights regions where trend chasing outperforms niche consistency and vice versa (Appendix A10).

Fig 11 — 95% Confidence Intervals for All Key Metrics

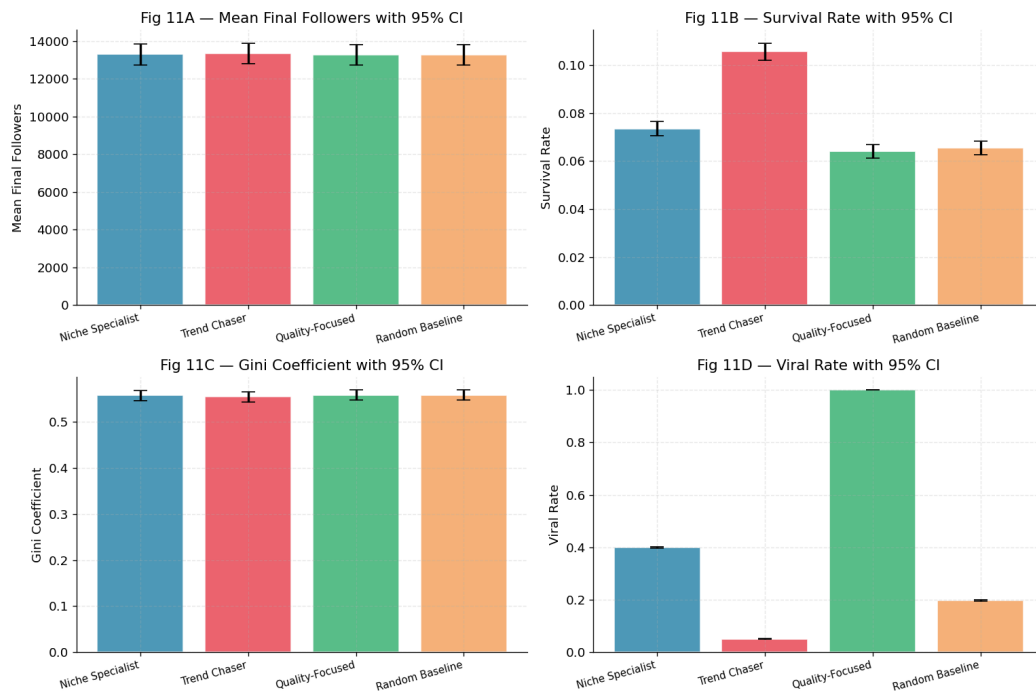


Figure 12: **Confidence intervals summarize what is statistically distinguishable.** Viral rates separate strategies cleanly; survival rates separate trend chaser from the rest; mean final followers and Gini overlap across strategies (Appendix A11).